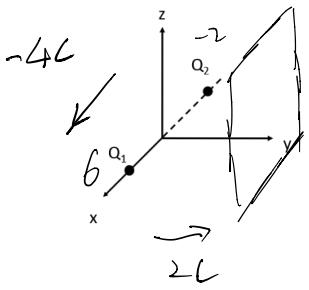
1. Two parts of this question are independent:

   a) (10 pts) Two unknown charges, $Q_1$ and $Q_2$ are located at $(x, y, z)$ =(1, 0, 0) and (-1, 0, 0), respectively, as shown below. The displacement flux $\int_{yz-plane} \mathbf{D} \cdot \hat{x} dydz$ through the entire $yz$-plane in the $+\hat{x}$ direction is -4 C. The flux through the plane $y = 2$ in the $+\hat{y}$ direction is 2 C. Determine $Q_1$ and $Q_2$ after writing a pair of algebraic equations relating the above displacement fluxes to $Q_1$ and $Q_2$.




$$\begin{cases} -4C = \frac{Q_2}{2} - \frac{Q_1}{2} \\ 2C = \frac{Q_2}{2} + \frac{Q_1}{2} \end{cases}$$

$$\Rightarrow \quad Q_1 = 6C$$
$$Q_2 = -2C$$

   b) Consider the electric field vector

$$\mathbf{E} = \hat{x} \sin y - \hat{y} \cos y \, \frac{\mathrm{V}}{\mathrm{m}}.$$

   i. (5 pts) Determine $\nabla \cdot \mathbf{E}$ and $\nabla \times \mathbf{E}$.

$$\nabla \cdot E = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \frac{d \sin y}{dx} + \frac{d - \cos y}{dy} + 0 = \sin y$$

$$\nabla \times E = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin y & -\cos y & 0 \end{vmatrix} = \hat{x} \cdot 0 - \hat{y} \cdot 0 + \hat{z} \cdot (\cos y) = \hat{z} \cos y$$

   ii. (5 pts) Which component of $\mathbf{E}$ is electrostatic? Explain,

$E_y = -\cos y$ is electrostatic.

$$\nabla \times (-\hat{y} \cos y) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & -\cos y & 0 \end{vmatrix} = 0$$

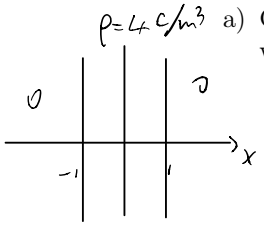
whereas $\nabla \times (\hat{x} \sin y) = \hat{z} \cos y \neq 0$

   iii. (5 pts) What is the static charge density $\rho$ implied by $\mathbf{E}$?

$$\nabla \cdot E = \rho / \varepsilon_0$$

$$\rho = \sin y \cdot \varepsilon_0 \quad C/m^3$$

2. Two parts of the following question are independent:




$4\,\text{C/m}^3 \cdot 2\,\text{m}$

$\rho_s = 8\ \text{C/m}^2$

a) Consider an infinite slab of width $W = 2$ m occupying the region $|x| < 1$ m. The charge density within the slab is $\rho = 4$ C/m$^3$ and it is zero outside the slab.

i. (4 pts) Determine the electric field vector $\mathbf{E}(x, 0, 0)$ for $x = 2$ m and $x = -2$ m.

$$\bar{E}(2,0,0) = \hat{x}\,\frac{8}{2\varepsilon_0} = \hat{x}\,\frac{4}{\varepsilon_0}\ \frac{V}{m}$$

$$\bar{E}(-2,0,0) = -\hat{x}\,\frac{8}{2\varepsilon_0} = -\hat{x}\,\frac{4}{\varepsilon_0}\ \frac{V}{m}.$$

ii. (4 pts) Determine the electric field vector $\mathbf{E}(x, 0, 0)$ for $-1 < x < 1$ m within the slab.

$$\bar{E} = \hat{x}\,\frac{\rho x}{\varepsilon_0} = \hat{x}\,\frac{4x}{\varepsilon_0}\ \frac{V}{m}.$$

iii. (2 pts) What is $\mathbf{E} \cdot d\mathbf{l}$ within the slab?

$$\bar{E}\cdot d\bar{l} = (E_x\hat{x} + E_y\hat{y} + E_z\hat{z})\cdot(\hat{x}dx + \hat{y}dy + \hat{z}dz)$$
$$= \hat{x}\,\frac{4x}{\varepsilon_0}\cdot\hat{x}\,dx$$
$$= \frac{4}{\varepsilon_0}x\,dx$$

iv. (5 pts) What is the electrostatic potential $V(x, y, z)$ within the slab if $V(0, 0, 0) = 0$ V?

$$\bar{E}\cdot d\bar{l} = -dV$$
$$\Rightarrow V = -\frac{2}{\varepsilon_0}x^2 + C$$

$$V(0,0,0) = 0 \Rightarrow C = 0$$

$$\boxed{V(x,y,z) = -\frac{2}{\varepsilon_0}x^2 \quad V}$$

b) (10 pts) Given that $\mathbf{E} = 2x\,\hat{x} - 4\sin(z)\,\hat{z}$ V/m, determine the electrostatic potential $V(x, y, z)$ if $V(0, 0, 0) = 4$ V.

$$\bar{E}\cdot d\bar{l} = 2x\,\hat{x}\cdot\hat{x}\,dx + (-4\sin(z))\,\hat{z}\cdot\hat{z}\,dz$$
$$= 2x\,dx - 4\sin(z)\,dz$$
$$= -dV$$

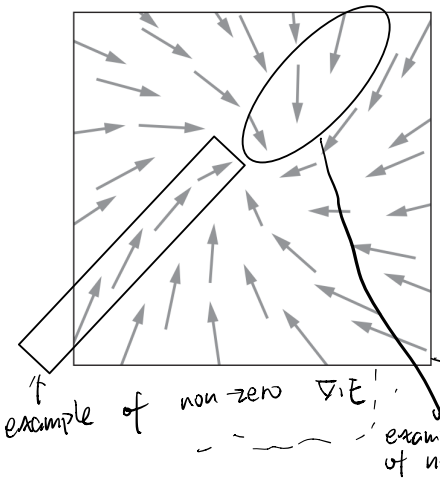
$$V = -x^2 - 4\cos(z) + C$$

$$V(0,0,0) = 4 \Rightarrow 4 = -0 - 4\cos(0) + C$$
$$C = 4 + 4\cos(0)$$
$$C = 8$$

$$V(x,y,z) = -x^2 - 4\cos(z) + 8\ \text{V}$$

3. Two parts of the following question are independent:

a) The 2-D vector field $\mathbf{E}$ depicted below has non zero $\nabla \cdot \mathbf{E}$ as well as $\nabla \times \mathbf{E}$.




i. (5 pts) Which feature of the field is an indicator of non-zero $\nabla \cdot \mathbf{E}$ — explain in a sentence or two.

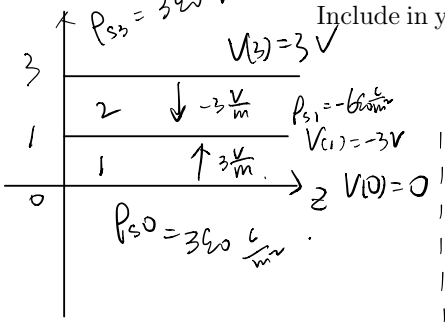
the magnitude of the vectors in the vector field is changing along their direction. The magnitude is becoming smaller and smaller as the vectors approaching the center.

ii. (5 pts) Which feature of the field is an indicator of non-zero $\nabla \times \mathbf{E}$ — explain in a sentence or two.

the magnitude of the vectors in the field is changing across their direction. the direction of the vectors in the circle is roughly parrallel, and their magnitude is increasing from left to right

b) Conducting plates placed on $z = 0$ and $z = 3$ m surfaces (of infinite extent) are separated by vacuum and maintained at potentials of $V(0) = 0$ V and $V(3) = 3$ V. Also, the potential of the $z = 1$ m surface that supports some surface charge density $\rho_{s1}$ is $V(1) = -3$ V.

i. (5 pts) What are the electric field vectors $\mathbf{E}_1$ and $\mathbf{E}_2$ below and above the $z = 1$ m surface? Include in your answer an appropriate diagram where $\mathbf{E}_1$ and $\mathbf{E}_2$ vectors are shown explicitly.




$V_2 = 3z$ V

$V_1 = -3z$ V

$\Rightarrow$ $\bar{E}_2 = -3\hat{z}$ V/m

$\bar{E}_1 = 3\hat{z}$ V/m

ii. (5 pts) What are the surface charge densities $\rho_{s0}$ and $\rho_{s3}$ at $z = 0$ and $z = 3$ m, respectively? Justify your answers.

$D_2 = E_2\varepsilon_0 = -3\varepsilon_0 \hat{z}$ C/m²

$D_1 = E_1\varepsilon_0 = 3\varepsilon_0 \hat{z}$ C/m²

$\Rightarrow$

$\rho_{s0} = \hat{z}(D_1 - 0) = 3\varepsilon_0$ C/m²

$\rho_{s3} = \hat{z}(0 - D_2) = 3\varepsilon_0$ C/m²

iii. (5 pts) What is the surface charge density $\rho_{s1}$ at $z = 1$ m? Justify your answer.

$\rho_{s1} = \hat{z}(D_2 - D_1)$

$= -3\varepsilon_0 - 3\varepsilon_0$

$= -6\varepsilon_0$ C/m²